

BASMEDSecure: A Logistic Regression-Based Data Hiding Framework for Securing Patient Records

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Abstract—Ensuring the confidentiality and integrity of sensitive patient records is a critical challenge in modern healthcare. Among various data security techniques, steganography, which conceals data within digital media, has emerged as a prominent solution. However, conventional methods often fail to achieve a sufficient balance between high data embedding capacity and the preservation of diagnostic image quality, limiting their practical use in clinical settings. To address this, this paper introduces BASMEDSecure, a novel data hiding framework that utilizes a logistic regression model to classify pixels based on their intensity levels. An adaptive embedding algorithm then utilizes this classification to dynamically adjust the number of secret data bits hidden within each pixel's least significant bit. Experimental results demonstrate promising performance for the proposed BASMEDSecure, with a PSNR (expressing the quality of the stego image) ranging from 52.103 to 75.521 decibels (dB). The obtained SSIM value of 0.9999 highlights a close similarity between the cover and stego medical images, emphasizing the efficiency of BASMEDSecure in maintaining image integrity, which is crucial for accurate medical interpretations.

Keywords—steganography, logistic regression, securing IT infrastructure, information security, data hiding

I. INTRODUCTION

The need to secure information has become more critical than ever, given the widespread use of the internet as a primary medium for data exchange, despite its inherent security vulnerabilities [1]. Among various data security techniques, data hiding has proven to be an effective strategy for concealing the existence of data during transmission [2]. Under the umbrella of data hiding, steganography embeds sensitive information within digital media such as images, rendering it imperceptible to human vision [3]. Image-based steganography results in a stego image, which is a cover image that has been altered by embedding sensitive data using a specific steganographic algorithm [4]. However, a problem in steganography has always been to increase the quality of stego images while also increasing the amount of information that can be embedded [5], [6], [7]. Low imperceptibility meant unsafe data transmission, which contradicts the purpose of steganography, and low capacity meant a more wasteful need for cover media to transmit large amounts of data.

In the current era, steganography has undergone significant evolution. One example of its improvement is the increased similarity between the cover image and the related stego image [8]. Although the quality of created stego images has improved, one problem that has persisted to this day is the amount of data that can be concealed into an image [5]. The amount of data embedded in a cover image has always been correlated to the quality of the stego image. As the amount of data increases, the susceptibility of the hidden data tends to increase proportionally, therefore compromising its security and integrity [5].

Current medical data management systems primarily focus on handling radiological images and textual patient information. Ensuring the security and confidentiality of these sensitive records remains a significant challenge, particularly during transmission and storage [9]. Steganography promises to be an effective technique to address such issues by concealing patient data within medical images, such as those in DICOM format. However, applying steganography in medical contexts introduces strict requirements, including maintaining the integrity of medical image metadata to preserve diagnostic accuracy and ensure compliance with clinical standards. Additionally, steganographic methods aim to balance the imperceptibility and payload capacity with competitive computational complexity. These challenges underscore the need for more adaptive and intelligent embedding mechanisms, as conventional methods often employ uniform strategies that fail to meet these competing demands efficiently.

Motivated by the gap in the state-of-the-art, this study proposes BASMEDSecure. This innovative steganographic approach utilizes a logistic regression algorithm to predict and classify pixels from grayscale cover images into three groups based on their intensity level. Logistic regression algorithm is chosen over existing machine learning paradigms due to its use for the basic task of identifying sensitive pixels in a cover image. This classification guides a dynamic embedding process where the number of embedded bits is adjusted based on the visual sensitivity of each pixel. Specifically, the process replaces up to three least significant bits (LSBs) of pixels with the lowest intensity and only one LSB for pixels with the highest intensity. By adaptively embedding data, this approach ensures the visual quality of the stego image remains

largely unaffected while significantly increasing the maximum embeddable capacity, thus enhancing both the efficiency and security of hidden data transmission. The main contributions of this study are threefold as follows:

- This study employs a logistic regression-based algorithm to predict and classify pixel intensities in grayscale cover images. By utilizing this prediction, the Method enables the embedding of secret data into every pixel, significantly increasing the overall payload capacity.
- An adaptive algorithm is proposed to dynamically determine the most suitable pixels for data hiding based on their intensity level. This approach ensures a balanced trade-off between imperceptibility and data capacity.
- The use of machine learning enables the identification of less sensitive regions within the image, which are more tolerant to modification. This targeted selection reduces the visual impact of data embedding and enhances steganographic security.

The following sections of this paper are arranged in four sections. Section II provides a detailed survey of the existing literature, and Section III outlines the proposed BASMEDSecure approach for adaptively concealing data. The last two sections, namely, Sections IV and V, present the experimentally obtained results and conclude the work, respectively.

II. LITERATURE SURVEY

Over the years, digital image steganography has emerged as a prominent technique for concealing sensitive information within visual media [1]. Extensive research has led to the development of various strategies aimed at maximizing the quality of concealment while preserving the integrity of the cover image [10], [11], [12], [13], [14]. These efforts have contributed to the advancement of sophisticated embedding techniques that enhance both consistency and imperceptibility. Moreover, several of the proposed methods actively address contemporary challenges, including detectability, data integrity, and overall reliability [15], [16], [17], [18], [19].

The Method proposed in [13] presents a steganography algorithm that secures high embedding capacity using LSB indicators. It employs a strategy of bit embedding that minimizes the amount of modified LSB in a cover image. Another method on [14]'s networking platforms. This Method enhances robustness through a preprocessing stage, an adaptive embedding strategy, and a postprocessing stage to handle distortions caused by data embedding. In [10], a method of steganography utilising Huffman Coding is introduced. They employed a novel method of stego image segmentation to mitigate concentration-induced distortions and optimized data embedding using Huffman Coding, thereby minimizing distortion while maximizing payload capacity. While these algorithms are able to create high quality stego images, they slightly lack in payload capacity and/or data extraction consistency.

Furthermore, in the research of [17], an image steganography strategy is introduced based on pixel topology and thresholding on pixel differences. The proposed Method employs a predetermined threshold that utilizes the RGB values of a cover image, converting them into binary data to implement the LSB replacement technique. The technique

presented by [19] is a novel approach to data hiding using image steganography based on color sensitivity. This technique is a unique application of the simple LSB replacement method, where pixels are selected in a spiral-like pattern, starting from the edges of the image and moving inward toward its center. Alternatively, a generative image steganography method utilising attribute feature transformation and mapping rules has been introduced in [18]. This approach disentangles a reference image to obtain content features and attribute features used for generating stego images. These methods demonstrate high performance in creating stego images, but are too complex for the amount of embeddable secret data in each image, such as the limited payload capacity in [18].

Additionally, some methods utilize deep learning algorithms to advance the field of steganography further. One such example is the Method proposed in [15], a generative image steganography technique similar to [18], which is based on deep learning. The paper examines the scheme of generative image steganography and then proposes a novel scheme based on deep learning. Next, the work in [16] introduces the use of deep neural networks in image steganography. The Method utilizes a stacked autoencoder comprising a preparation, hidden, and reveal networks to embed an encrypted secret image into a cover image and subsequently extract it. However, these methods have relatively low payload capacity, significant image degradation, and high computational cost due to the use of deep learning models.

Steganography has also found important applications in the medical field, where the secure handling of sensitive patient information is paramount. The work in [11] presents a method for using steganography to conceal medical reports within DICOM images. This strategy decomposes the medical cover image with an integer wavelet filter, before being injected with an already compressed version of patients' confidential information. Additionally, the Method proposed in [12] helps ensure medical data privacy in digital images by utilizing the contour principle within a steganographic scheme. The presented Method applies contour detection, combined with K-means clustering, to segment pixel groups across the RGB image layers and then uses LSB substitution to embed encrypted medical data into the selected pixel clusters. Although these strategies demonstrate strong imperceptibility and adequate data concealment, they still suffer from low payload capacity, high computational complexity, and vulnerability to steganalysis methods.

The survey of these diverse approaches reveals that, despite considerable progress, the issue of finding an optimal balance between embedding capacity and imperceptibility remains. Adapting to diverse and unique image characteristics is often a challenge for current methods, resulting in reduced efficiency. This gap demonstrates the need for a more adaptive embedding mechanism. Hence, the paper proposes BASMEDSecure, a novel steganographic scheme that adaptively hides sensitive patient records into medical images.

III. METHODOLOGY

BASMEDSecure presents an adaptive method for using the logistic regression algorithm to predict pixel intensities from medical DICOM images. A dynamic embedding algorithm selectively hides secret data within labeled pixels, where the number of bits to be hidden in each pixel is

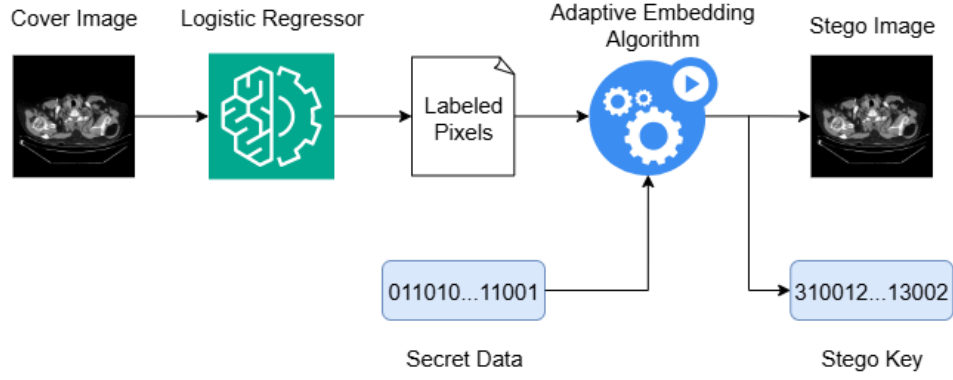


Fig. 1. Overall Logic of the Approach in BASMEDSecure

determined by its intensity (Low (L), Medium (M), and High (H)). All data are embedded only into the least significant bits of labeled pixels. The flowchart in Fig. 1 illustrates the overall steps involved in the BASEMEDSecure steganographic embedding process.

A. Data Embedding

The proposed steganographic Method applies a dynamic embedding algorithm to hide secret data within the cover DICOM image pixels by replacing pixel bits with secret data in the form of bits. To minimize changes to the cover image, secret bits would only be replacing the least significant bits of each pixel (up to three). The number of bits hidden in each pixel is determined by its classified intensity level and total secret data length, with the following cases:

- 1) *Case 1: Data length $\leq P_0$* : Embedding one bit in each low intensity pixel.
- 2) *Case 2: Data length $\leq P_0 * 2 + P_1$* : Embedding two bits in each low intensity pixel and one bit in each medium intensity pixel.
- 3) *Case 3: Data length $\geq P_0 * 2 + P_1$* : Embedding three bits in each low intensity pixel, two in each medium intensity pixel, and one in each high intensity pixel.

This proposed Method allows for a maximum payload capacity, as mathematically expressed in (1). Considering P_0 , P_1 , and P_2 represent the number of pixels predicted to be low intensity, medium intensity, and high intensity, respectively. During this embedding process, the number of bits to embed in each pixel is stored in an array of the exact dimensions as the cover image, serving as a data key for extraction. Once all data is embedded or all pixels used, a steganography image is generated with the same dimensions and format as the cover image. The complete adaptive data embedding process is outlined in the pseudocode presented in Algorithm I. It is important to note that the proposed algorithm uses logistic regression to identify pixels based on their intensities adaptively. The grayscale pixels are extracted from TIFF images and labeled with integer values representing their intensity levels: 0 for low intensity (0–100), 1 for medium intensity (101–150), and 2 for high intensity (151–255). These labeled values serve as target classes for training a logistic regression model, which learns to classify pixels based on their intensity. Once trained, the model enables automatic pixel grouping during the embedding process, allowing the

system to adaptively determine embedding depths according to the visual intensity of each region.

$$Capacity = 3P_0 + 2P_1 + 1P_2 \quad (1)$$

Algorithm I Pseudocode of the Embedding Process

```

1) Input: Cover Image, Secret Data
2) Read Cover Image and Secret Data
3) Predict pixel intensity, resulting in Pixels and Labels
4) If length of Secret Data  $\leq$  (amount of Labels[0]):
5)   Repeat
6)     Read a Pixels entry
7)     If Pixels entry's label = 0:
8)       Replace 1 LSB with 1 bit of Secret Data
9)       Increment Secret Data index by 1
10)      Append a key of value = 1
11)      Increment Pixels index by 1
12)    Until all Secret Data has been embedded
13)  Else if length of Secret Data  $\leq 2 * (\text{amount of Labels}[0]) + (\text{amount of Labels}[1])$ :
14)    Repeat
15)      Read a Pixels entry
16)      If Pixels entry's label = 0:
17)        Replace 2 LSBs with 2 bits of Secret Data
18)        Increment Secret Data index by 2
19)        Append a key of value = 2
20)      Else if Pixels entry's label = 1:
21)        Replace 1 LSB with 1 bit of Secret Data
22)        Increment Secret Data index by 1
23)        Append a key of value = 1
24)      Increment Pixels index by 1
25)    Until all Secret Data has been embedded
26)  Else:
27)    Repeat
28)      Read a Pixels entry
29)      If Pixels entry's label = 0:
30)        Replace 3 LSBs with 3 bits of Secret Data
31)        Increment Secret Data index by 3
32)        Append a key of value = 3
33)      Else if Pixels entry's label = 1:
34)        Replace 2 LSBs with 2 bit of Secret Data
35)        Increment Secret Data index by 2
36)        Append a key of value = 2
37)      Else if Pixels entry's label = 2:
38)        Replace 1 LSB with 1 bit of Secret Data
39)        Increment Secret Data index by 1
40)        Append a key of value = 1
41)      Increment Pixels index by 1
42)    Until all Secret Data has been embedded
43) Output: Stego Image, Key Array

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Algorithm II Detailed Steps for the Data and Extraction

- 1) **Input:** Stego Image, Key Array
- 2) Read Stego Image
- 3) Predict pixel intensity, resulting in Pixels and Labels
- 4) If length of Secret Data $\leq$ (amount of Labels[0]):
- 5) Repeat
- 6) Get an entry of Key Array
- 7) If Key Array's entry = 1:
- 8) Read corresponding pixel entry of Stego Image
- 9) Extract 1 bit from the pixel's LSBs
- 10) Append bit to list of Secret Bits
- 11) If Key Array's entry = 2:
- 12) Read corresponding pixel entry of Stego Image
- 13) Extract 2 bits from the pixel's LSBs
- 14) Append bits to list of Secret Bits
- 15) If Key Array's entry = 3:
- 16) Read corresponding pixel entry of Stego Image
- 17) Extract 3 bits from the pixel's LSBs
- 18) Append bits to list of Secret Bits
- 19) Increment Key Array's index by 1
- 20) Until Key Array's index > length of Key Array
- 21) **Output:** Secret Bits, Cover image

B. Extraction Process

Based on the primary objective of steganography, to conceal data from unauthorized access, a corresponding, discreet extraction mechanism must be employed to retrieve the hidden information. In the proposed BASMEDSecure, the data extraction is performed using a straightforward LSB extraction process, where the previously generated stego key guides the number of bits retrieved from each pixel. The hidden data is sequentially reconstructed by traversing the image from the first pixel to the last. The detailed extraction procedure is outlined in Algorithm II, which contains all steps taken from input to output.

IV. RESULTS AND DISCUSSIONS

This section presents the experimental results obtained from the proposed BASMEDSecure steganographic method. Performance is quantitatively measured using two key image quality metrics: Peak Signal-to-Noise Ratio (PSNR) and

Structural Similarity Index Measure (SSIM). PSNR is mathematically computed using (2), and it is a widely used metric that quantifies the ratio between the maximum possible power of a signal and the power of noise that affects its fidelity. To calculate the SSIM, the relation in (3) is used, and this aims to evaluate the perceptual similarity between two images by comparing their luminance, contrast, and structural information. The variable C_1 represents the cover image, and C_2 represents the stego image, μ_x and μ_y represent the mean pixel intensity value of the cover image and stego image, respectively, σ_x and σ_y represent the variance of pixel intensities of the cover image and stego image, respectively, and σ_{xy} represents the covariance of the pixel intensities between the two images. It is also crucial to note that the MAX_I is equivalent to 255, the maximum possible pixel intensity that is possible to be seen. The evaluation is conducted on a DICOM image dataset [20], images from the Massachusetts Infectious Disease Hospital library [21], and a collection of CT images from [22]. The experimental secret bits are generated from the Lorem Ipsum story [23]. It is important to note that for experimentation we consider individual image of size 512×512 , meaning that the regression based identification of embeddable pixels considers 262 144 pixels as input classification data.

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{\frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i,j) - K(i,j)]^2} \right) \quad (2)$$

$$SSIM = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3)$$

A. Experimental Results

The experimental results in Tables I and II demonstrate the effectiveness and robustness of the proposed BASMEDSecure across various DICOM images, including anatomical and anonymized samples. As shown in Table I, the PSNR values start at approximately 75.43 dB for all cover images at the lowest payload of 1 kb, indicating almost negligible distortion. As the payload increases to 100 kb, the PSNR gradually decreases but remains within acceptable quality bounds. For instance, even at 100 kb, images such as Abdominal, Hand, and Head retain PSNR values above 55.3 dB, while Chest reaches 52.1 dB, which still indicates high

TABLE I. OBTAINED PSNR RESULTS FROM EXPERIMENTATION

Cover Image	Payload Size in kb										
	1	10	20	30	40	50	60	70	80	90	100
Abdominal	75.432	65.275	62.313	60.583	59.284	58.338	57.537	56.882	56.322	55.787	55.329
Chest	75.387	65.313	62.327	60.579	59.293	58.336	54.113	53.645	53.052	52.552	52.103
Hand	75.432	65.272	62.331	60.569	59.294	58.341	57.546	56.886	56.307	55.754	55.332
Head	75.432	65.264	62.306	60.547	59.319	58.314	57.521	56.866	56.304	55.786	55.31
Anonymized 1	75.432	65.275	62.313	60.585	59.285	58.336	57.523	56.874	56.327	55.776	55.328
Anonymized 2	75.432	65.275	62.313	60.585	59.285	58.336	57.523	56.874	56.328	55.773	55.329
Anonymized 3	75.432	65.275	62.313	60.585	59.285	58.337	57.518	56.863	56.337	55.803	55.325
Anonymized 4	75.432	65.275	62.313	60.585	59.285	58.336	57.532	56.867	56.334	55.783	55.335
Anonymized 5	75.432	65.275	62.313	60.585	59.289	58.334	57.527	56.881	56.315	55.794	55.325
Anonymized 6	75.432	65.275	62.313	60.585	59.285	58.33	57.541	56.874	56.34	55.785	55.321
Anonymized 7	75.432	65.275	62.313	60.585	59.29	58.347	57.523	56.87	56.313	55.792	55.321
Anonymized 8	75.432	65.275	62.313	60.585	59.285	58.33	57.507	56.887	56.324	55.8	55.336
Anonymized 9	75.432	65.275	62.313	60.585	59.285	58.331	57.532	56.861	56.33	55.795	55.34
image-00000	75.432	65.275	62.313	60.585	59.285	58.336	57.522	56.867	56.318	55.788	55.336
image-00001	75.423	65.274	62.319	60.565	59.299	58.348	57.535	56.875	56.288	55.782	55.32
image-00002	75.432	65.275	62.313	60.585	59.291	58.336	57.535	56.865	56.312	55.772	55.332
image-00003	75.432	65.275	62.313	60.585	59.285	58.336	57.523	56.874	56.327	55.773	55.324
image-00004	75.521	65.313	62.333	60.584	59.293	58.327	57.521	56.899	56.318	55.77	55.326

TABLE II. EXPERIMENTAL RESULTS IN SSIM

Cover Image	Payload Size in kb										
	1	10	20	30	40	50	60	70	80	90	100
Abdominal	0.9999	0.9989	0.9975	0.9959	0.9947	0.9938	0.9928	0.9919	0.9912	0.9901	0.9891
Chest	0.9999	0.9998	0.9996	0.9994	0.9993	0.9992	0.9979	0.9976	0.9972	0.997	0.9967
Hand	0.9999	0.9989	0.9978	0.9967	0.9955	0.9945	0.9935	0.9927	0.992	0.9911	0.9906
Head	0.9999	0.9993	0.9991	0.9989	0.9986	0.9984	0.9983	0.9981	0.9979	0.9977	0.9975
Anonymized 1	0.9999	0.9986	0.997	0.9955	0.9938	0.9922	0.9905	0.989	0.9876	0.9861	0.9856
Anonymized 2	0.9999	0.9986	0.997	0.9955	0.9938	0.9922	0.9906	0.9896	0.9888	0.9875	0.9864
Anonymized 3	0.9999	0.9986	0.997	0.9955	0.9938	0.9922	0.991	0.9901	0.9893	0.9881	0.987
Anonymized 4	0.9999	0.9986	0.997	0.9955	0.9938	0.9923	0.9913	0.9905	0.9895	0.9883	0.9872
Anonymized 5	0.9999	0.9986	0.997	0.9955	0.9943	0.9935	0.9927	0.9921	0.9916	0.9908	0.9902
Anonymized 6	0.9999	0.9986	0.997	0.9955	0.9943	0.9935	0.9928	0.9921	0.9917	0.991	0.9905
Anonymized 7	0.9999	0.9986	0.997	0.9955	0.9943	0.9935	0.9928	0.9921	0.9917	0.991	0.9905
Anonymized 8	0.9999	0.9986	0.997	0.9955	0.9938	0.9928	0.9921	0.9915	0.9911	0.9905	0.9901
Anonymized 9	0.9999	0.9989	0.9975	0.9959	0.9947	0.9938	0.9928	0.9919	0.9912	0.9901	0.9891
image-00000	0.9999	0.9998	0.9996	0.9994	0.9993	0.9992	0.9979	0.9976	0.9972	0.997	0.9967
image-00001	0.9999	0.9989	0.9978	0.9967	0.9955	0.9945	0.9935	0.9927	0.992	0.9911	0.9906
image-00002	0.9999	0.9993	0.9991	0.9989	0.9986	0.9984	0.9983	0.9981	0.9979	0.9977	0.9975
image-00003	0.9999	0.9986	0.997	0.9955	0.9938	0.9922	0.9905	0.989	0.9876	0.9861	0.9856
image-00004	0.9999	0.9986	0.997	0.9955	0.9938	0.9922	0.9906	0.9896	0.9888	0.9875	0.9864

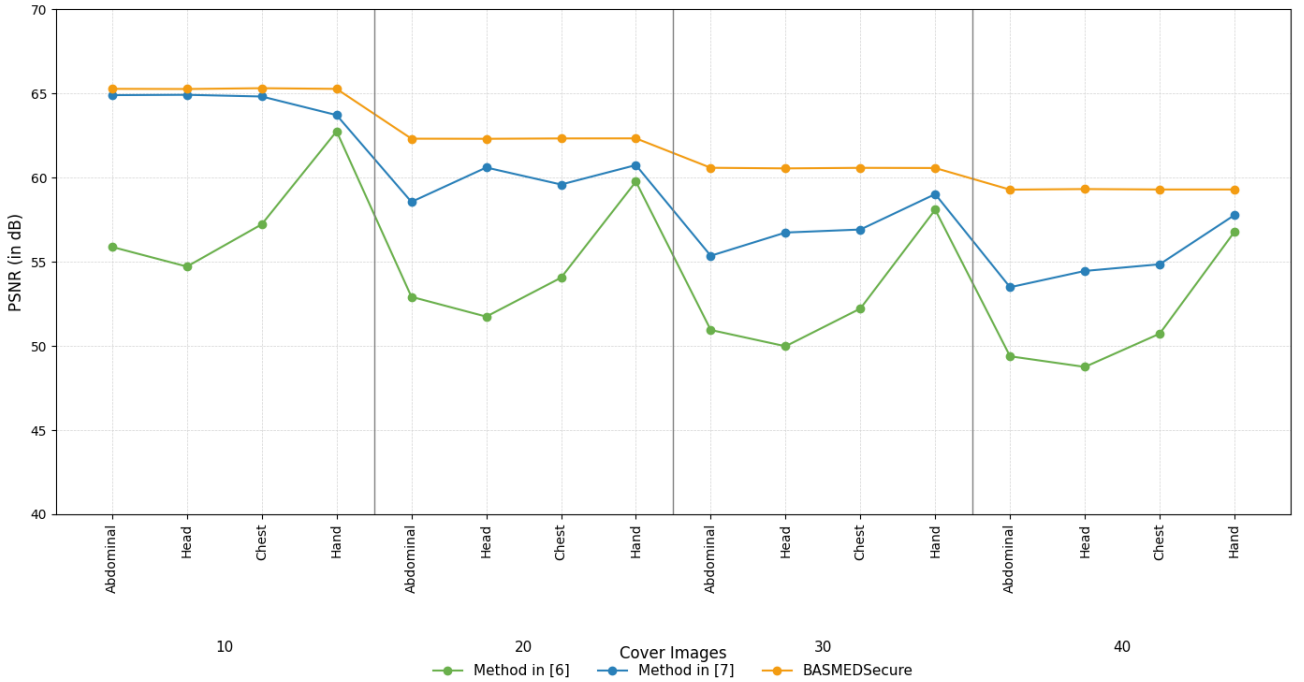


Fig. 2 PSNR Comparison to Existing Benchmarks

visual fidelity. This slow and controlled decline underscores the model's ability to maintain imperceptibility even under high embedding rates.

Table II further supports this observation through SSIM results, which assess perceptual similarity. At a 1 kb payload, all images yield near-perfect scores of 0.9999, indicating virtually no structural distortion. Even as the payload reaches 100 kb, SSIM values remain promising, demonstrating that the stego images closely resemble their original counterparts in terms of structure and texture. These high SSIM and PSNR scores across a diverse set of images confirm the generalizability of the proposed BASMEDSecure method and its suitability for securely embedding medical data while preserving both image quality and diagnostic value.

B. Results Comparison to the Existing Benchmarks

Fig. 2 presents a comparison of the PSNR performance among three different methods: Method in [6], the Method in

[7], and the proposed BASMEDSecure. PSNR, a key metric for evaluating the fidelity of image reconstruction, is considered for comparison because it reflects the quality of the stego medical image after embedding the patient records. Considering that higher PSNR values indicate better preservation of image quality, BASMEDSecure consistently outperforms both baseline methods, maintaining PSNR values above 60 dB, demonstrating superior robustness and minimal distortion regardless of image category or quantity. In contrast, the methods in [6] and [7] exhibit substantial variability, with noticeable degradation especially evident in Method [6] as the number of images increases.

V. CONCLUSION

Steganography has advanced considerably in the digital era, evolving into a powerful technique for embedding sensitive data within multimedia content. However, a core challenge persists in achieving a balance between data capacity and imperceptibility. This study introduces

BASMEDSecure, a steganographic framework tailored for securely embedding patient information in medical DICOM images. By incorporating logistic regression to classify pixel intensities, the Method enables adaptive and efficient data embedding across the entire image. The proposed approach demonstrates strong visual fidelity and structural integrity, maintaining the quality of the stego images while supporting secure and reliable information hiding.

For future work, the proposed BASMEDSecure framework can be extended by incorporating deep learning models to enhance pixel classification accuracy and embed decisions further. Additionally, integrating encryption prior to embedding may improve data confidentiality, especially for sensitive medical records.

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